standard candles: Objects of a known intrinsic brightness that are useful for measuring astronomical distances.

standard model: Quantum mechanical theory composed of quantum chromodynamics and the electroweak theory; describes all matter and forces, except for gravity. Based on conception of point particles.

strong nuclear force: Force of nature that influences quarks; holds quarks together inside protons and neutrons.

string theory: Theory based on one-dimensional vibrating filaments of energy (see superstring theory), but which does not necessarily incorporate supersymmetry. Sometimes used as shorthand for superstring theory.

superstring theory: Theory in which fundamental ingredients are one-dimensional loops (closed strings) or snippets (open strings) of vibrating energy, which unites general relativity and quantum mechanics; incorporates supersymmetry.

supersymmetry: A symmetry in which laws are unchanged when particles with a whole number amount of spin (force particles) are interchanged with particles that have half of a whole number amount of spin (matter particles).

symmetry: A transformation on a physical system that leaves the system's appearance unchanged (e.g., a rotation of a perfect sphere about its center leaves the sphere

unchanged); a transformation of a physical system that has no effect on the laws describing the system.

time-reversal symmetry: Property of the accepted laws of nature in which laws make no distinction between one direction in time and the other. From any given moment, the laws treat past and future in exactly the same way.

time slice: All of space at one moment of time; a single slice through the spacetime block or loaf.

translational invariance, translational symmetry: Property of accepted laws of nature in which the laws are applicable at any location in space.

uncertainty principle: Property of quantum mechanics in which there is a fundamental limit on how precisely certain complementary physical features can be measured or specified.

unified theory: A theory that describes all forces and all matter in a single theoretical structure.

vacuum: The emptiest that a region can be; the state of lowest energy.

vacuum field fluctuations: See *quantum fluctuations.* velocity: The speed and direction of an object's motion.

W and Z particles: The messenger particles of the weak nuclear force.

wavefunction: See *probability wave.*

weak nuclear force: Force of nature, acting on subatomic scales, and responsible for phenomena such as radioactive decay.